

Adaptation and Vertical Control in the Airline Industry

Silke Forbes & Mara Lederman

American Economic Review
(2009)

What Explains “Make” or “Buy”?

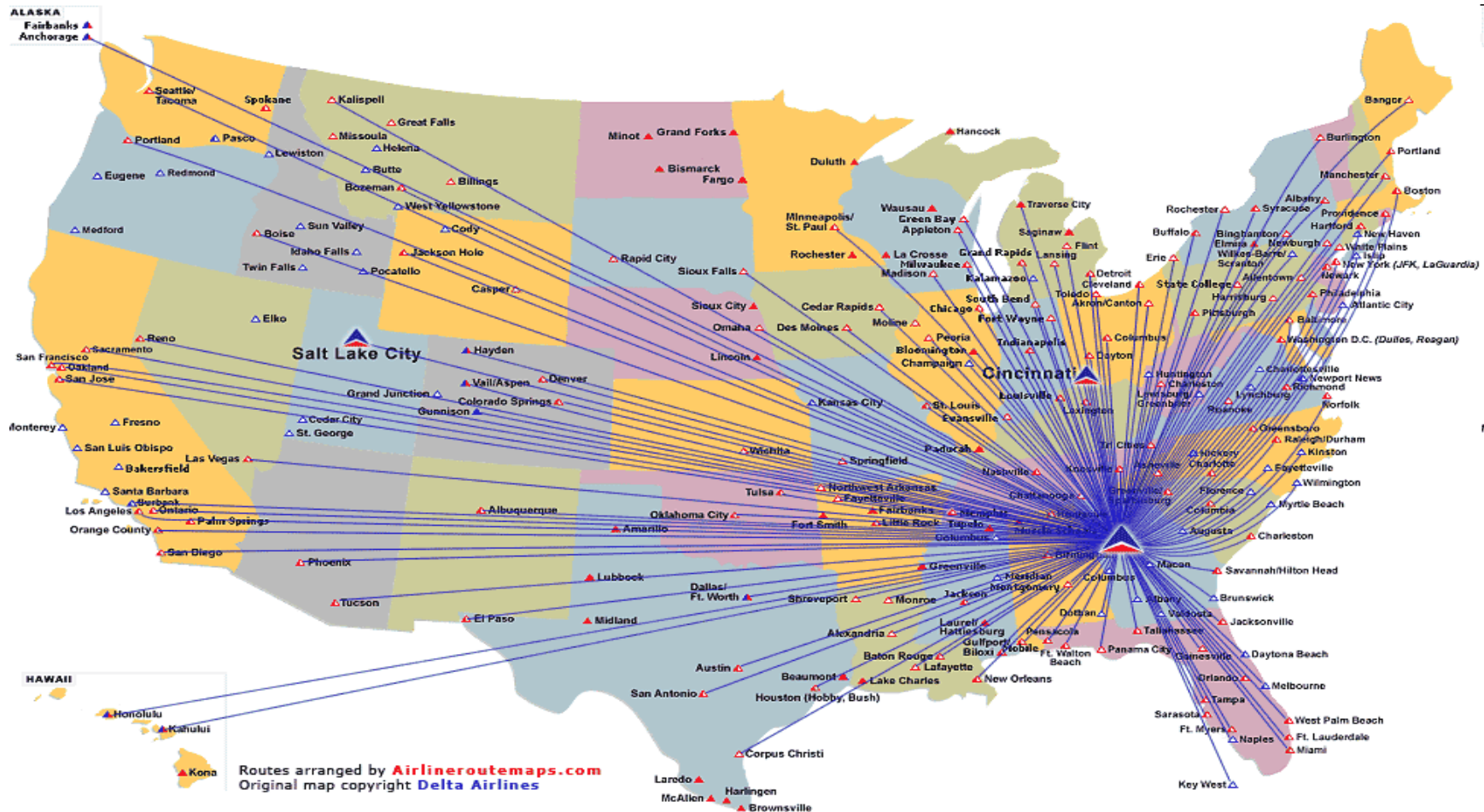
- So far: asset-specificity
 - E.g. mine-mouth coal mine and powerplant (Joskow)
- Another possibility
 - Transaction complexity over ex post adaptation decisions
 - What happens when the “production process” doesn’t go as planned?
 - If “buy” (on the spot market), there may be substantial transaction costs to adapting
 - ➔ incentive to vertically integrate

- What is the research question?
- What is the goal of the paper? (does it develop new methods, answer a policy question, test theoretical models, measure some effect, etc.?)
- Why is it important, according to the author?
- What are the data, exactly? What are the key variables?
- What is the most important empirical relationship in the paper (literally – not the author's interpretation of the empirical relationship)?
- What is the author's interpretation?
- Are there any other interpretations of the empirical results? Does the author consider or discuss these alternatives?
- Why does the author choose this particular reduced form model?
- Does the author discuss the underlying structural model that yields the particular reduced form model?
- Could imposing more structure answer additional questions of interest?

Additional questions about the data:

- What are the key variables in the empirical model?
- What is assumed to be exogenous and endogenous? Are there variables that are treated as exogenous that can conceivably be endogenous?
- What is the key variation in the data that is identifying the parameters of interest?
- Are there distributional assumptions that are driving identification?

Delta's Network Out of Atlanta



Source: http://www.airlineroutemaps.com/USA/Delta_Airlines.shtml

Good survey on airline industry in US: [How Airline Markets Work...Or Do They? Regulatory Reform in the Airline Industry,](#) Borenstein & Rose, NBER working paper #13452.

Delta Regional Partner: Atlantic Southeast Airlines



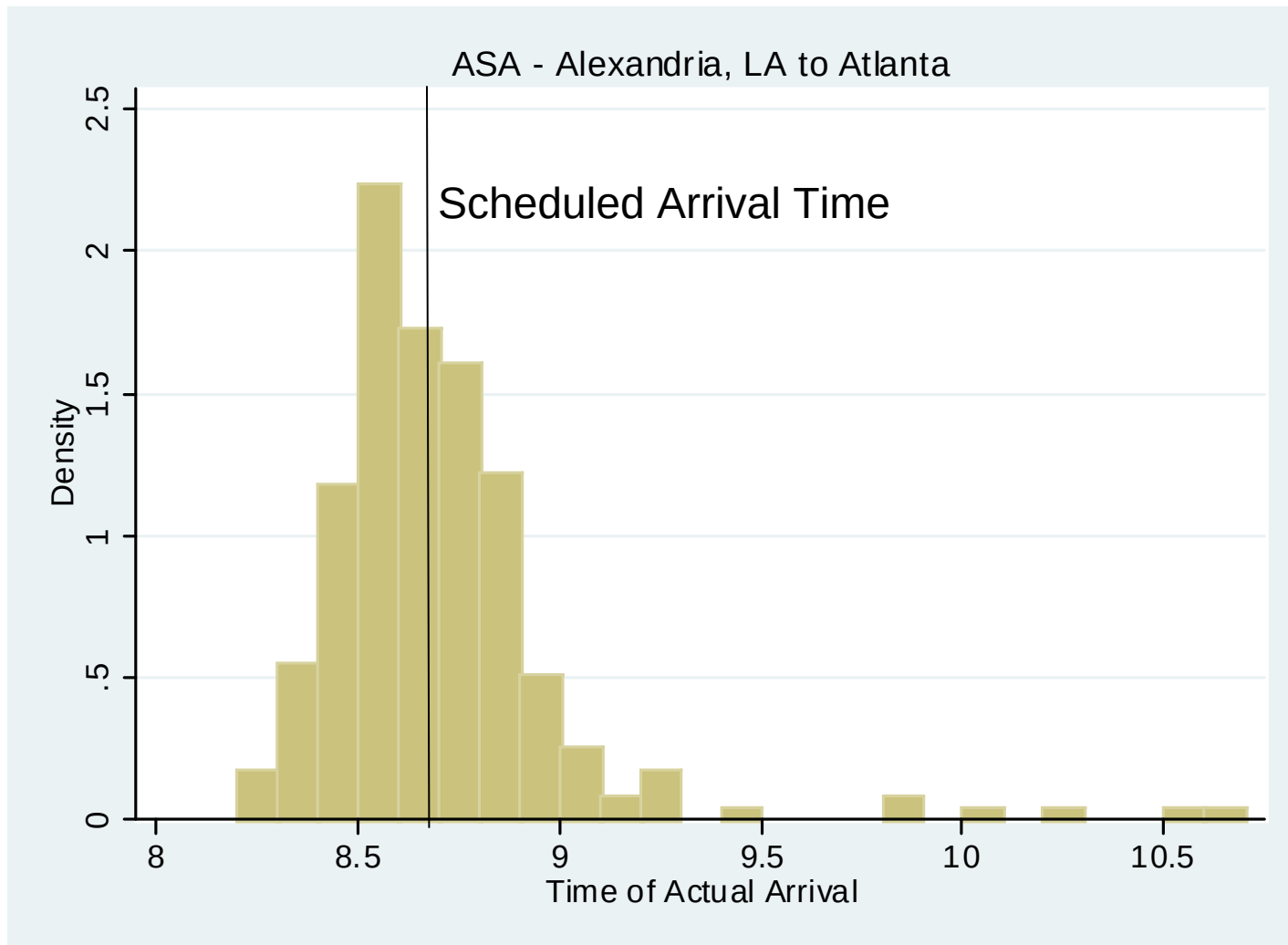
route system



Routes subject to change without notice

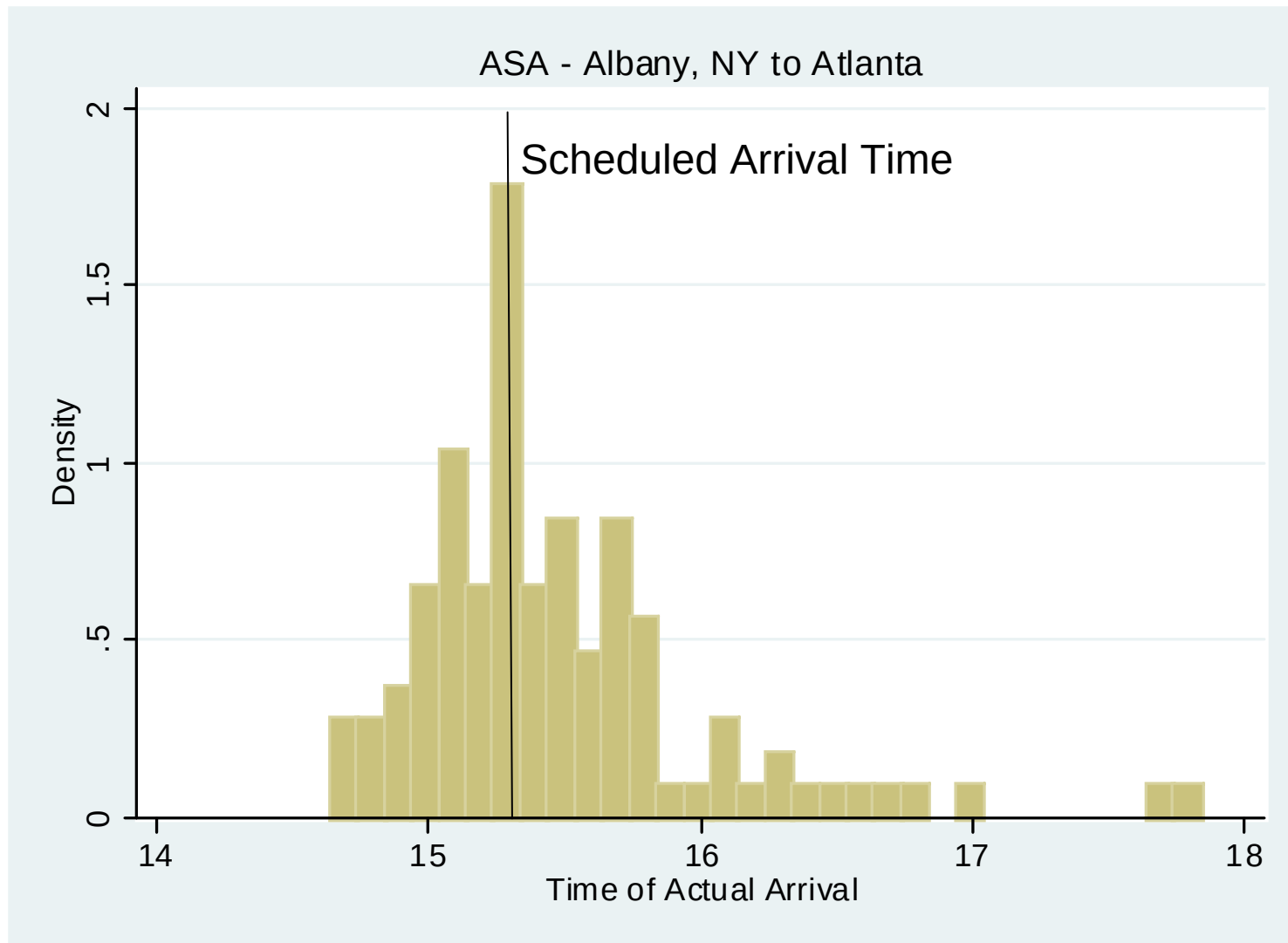
Shorter routes, connecting smaller cities (often to hub)

Does production “go as planned” in the airline industry?

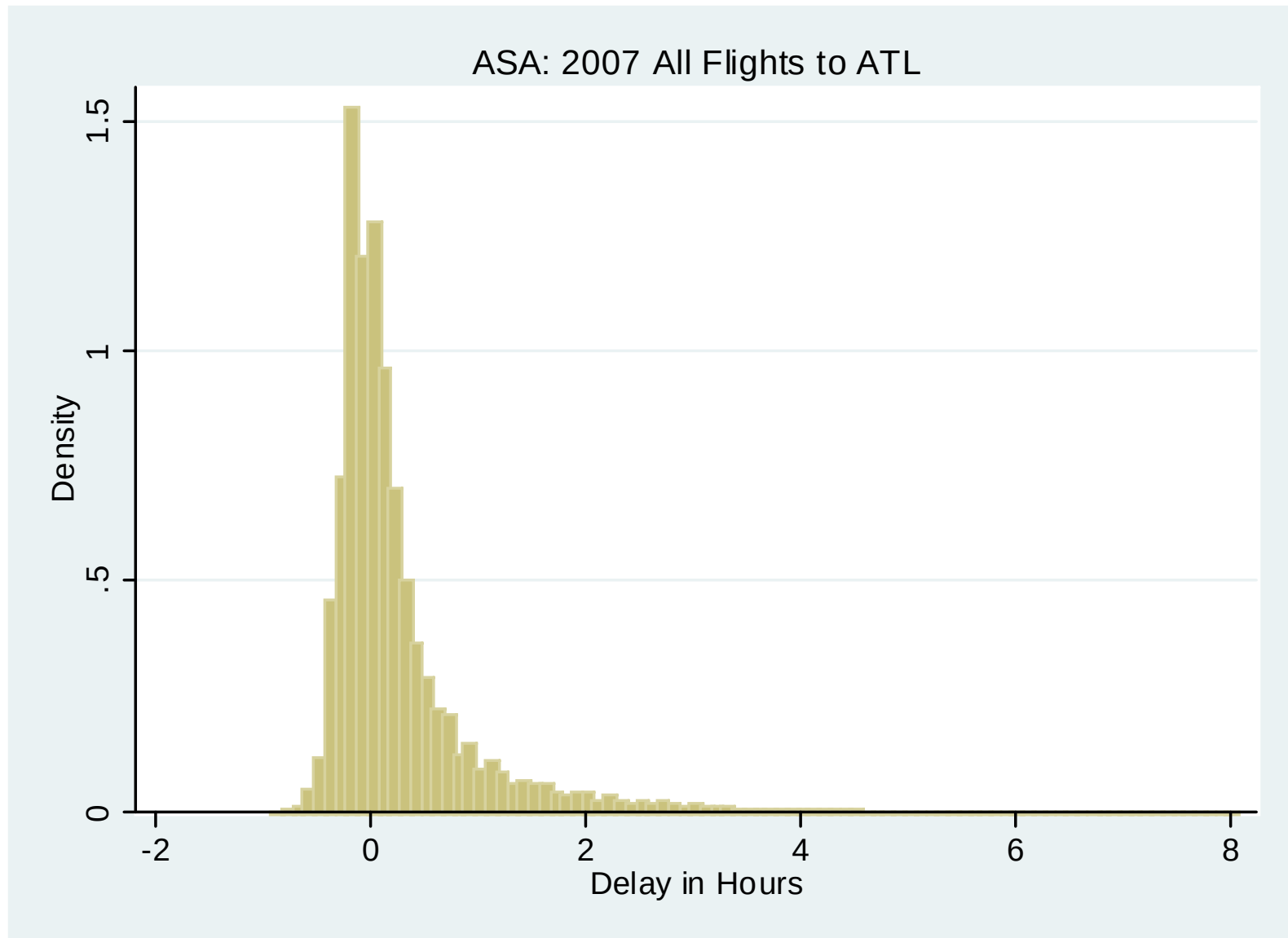


2007 Data available from: http://www.transtats.bts.gov/DL_SelectFields.asp?Table_ID=236&DB_Short_Name=On-Time

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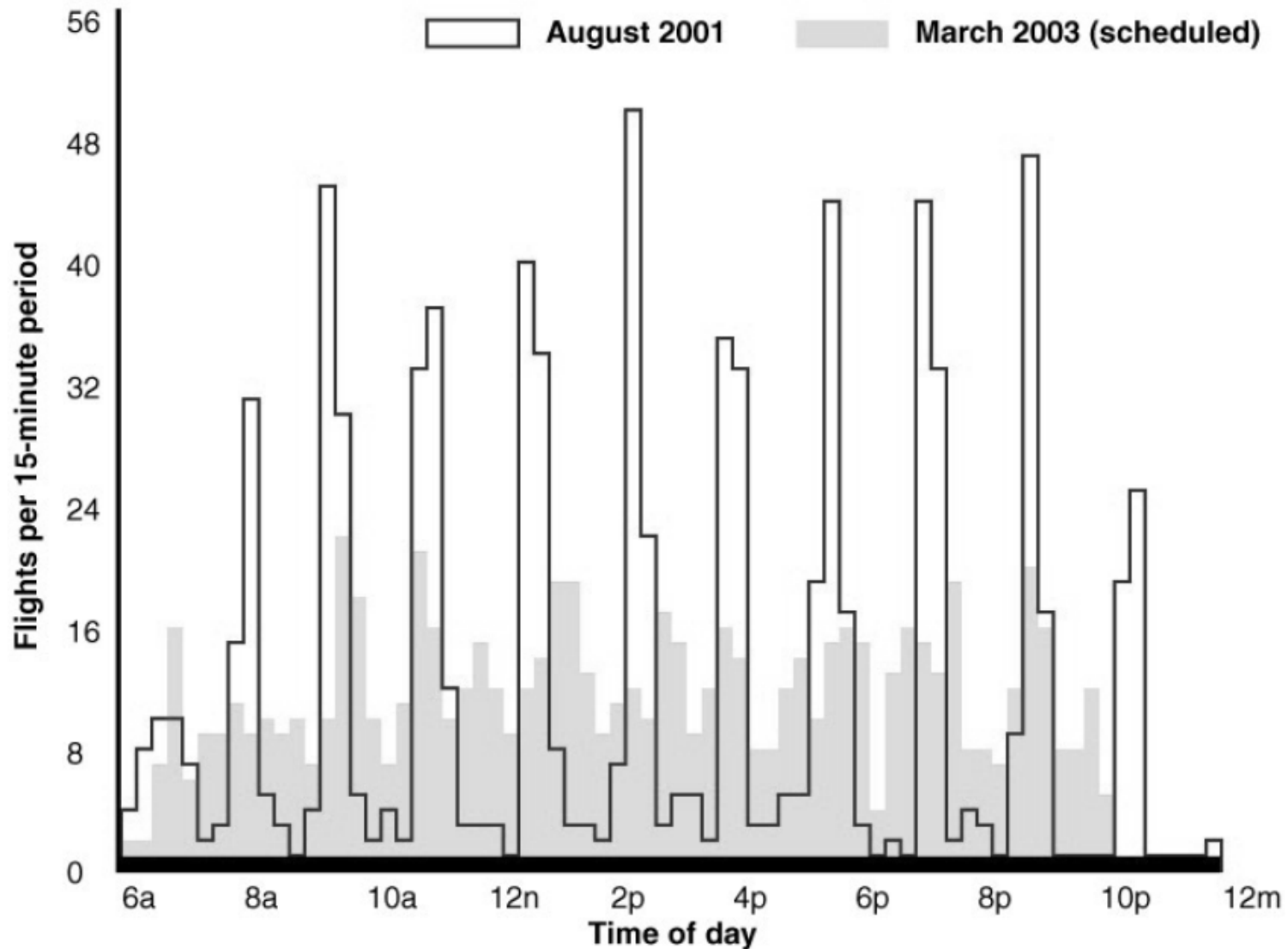


ASA: Delays on Flights to ATL



“Banking” at Hub Airports

American at DFW



Source: Borenstein & Rose (2007)

Visualization of How “Networked” the Industry Is

- See FlightPatternsHD.mp4

What Happens in Airlines When Delays Occur?

- Rationing of landings on runways, use of gates
- Outgoing flights held for connecting passengers on late incoming flights
- Outgoing flights delayed because the aircraft hasn't arrived
- Crew may be unavailable if limits on shift are reached
-

Incentives to Vertically Integrate

- Contracts are incomplete
 - Can contract on ex ante schedule
 - Cannot contract on ex post changes due e.g. to weather
 - “Best” change can depend on everything else in the network
 - Even if they could, it would be inefficient to remove flexibility
- Face different ex post incentives
 - Independent regionals care only about routes they serve; majors care about entire network
 - Haggling over the adjustments (especially in real-time) has substantial transaction costs

What Would Grossman-Hart Say?

- Just contractually allocate the control rights to the major
 - Apparently, not so easy to do
 - Ordering schedule changes is possible, but implementing those schedule changes still problematic
 - Major can't force a regional plane fly at a certain time, with a certain crew, etc.
 - G-H model assumes firm with control can determine precisely how asset is used

Incentives NOT to Integrate

- Regionals have lower costs
 - United: majors' senior pilots make 80% more than regionals, flight attendants 32% more, mechanics 31% more
 - Major may be unable to commit not to pay a regionals employees comparably

Table 1
Majors and Regional Partners in 2000

MAJOR	REGIONAL PARTNER
American Airlines	American Eagle Airlines Business Express
Continental Airlines	Continental Express Gulfstream International Airlines
Delta Air Lines	Atlantic Coast Airlines/ACJet Atlantic Southeast Airlines Comair SkyWest Airlines Trans States Airlines
Northwest Airlines	Express Airlines, I Mesaba Aviation
Trans World Airlines	Chautauqua Airlines Trans States Airlines
United Airlines	Air Wisconsin Atlantic Coast Airlines Great Lakes Aviation Gulfstream International Airlines SkyWest Airlines
US Airways	Mesa Air Group/Air Midwest Allegheny Airlines Mesa Air Group/CCAir Chautauqua Airlines Colgan Airways Commutair Mesa Air Group/Mesa Airlines Piedmont Airlines PSA Airlines

Regional carriers in bold are fully owned by the major

4 airlines that use
both independent and
owned regionals

Overall “Identification Strategy”

- Observe (exogenous?) variation in frequency of need to make ex post adaptation decisions
 1. Weather
 2. A route (city pair) that is more integrated into the airline’s network
 - Flight may contain connecting passengers
 - Flight competes for scarce resources such as landing slot, gate, ground crew, etc.

Major Airlines Choice Set for a Given City Pair (with a prob(adaptation))

1. Serve citypair itself
 2. Use regional that is owned
 3. Use regional that is independent
- Key Assumption (exogeneity):
 - None of explanatory variables are correlated with unobserved factors that affect benefits of either of 3 choices

Sample

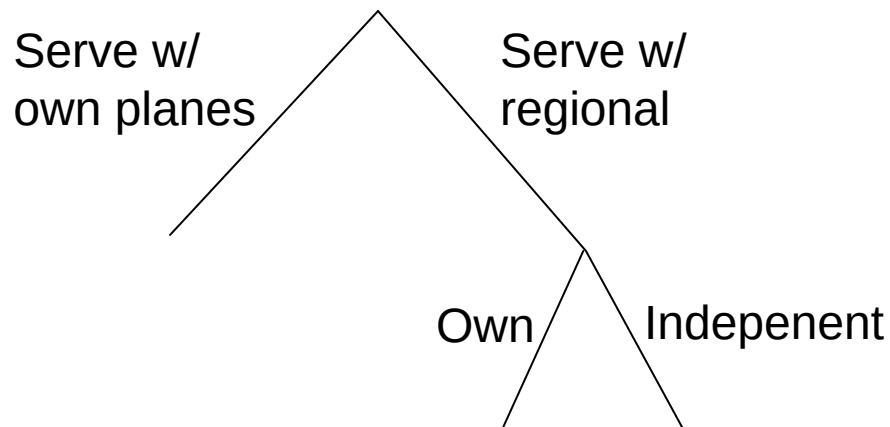
- 2000 Quarter 2
- Travel between largest 300 U.S. airports on flights ≤ 1500 miles (so regional can serve)
- “Weekday flights”
- Observation = airline-citypair
- ➔ 1742 airline-citypairs
 - 56% served by major, 26% by owned regionals, 19% indep regionals

Table 3
Summary Statistics

VARIABLE NAME	N	MEAN	ST. DEV.	MIN	MAX
<i>Regional</i>	1742	0.44	0.50	0	1
<i>Owned Regional, given Regional=1</i>	774	0.58	0.49	0	1
<i>Hub</i>	1742	0.76	0.43	0	1
<i>Number of Flights at Larger Endpoint</i> (in 00s)	1742	3.55	2.12	0.02	8.23
<i>Number of Flights at Smaller Endpoint</i> (in 00s)	1742	0.19	0.41	0	4.91
<i>Precipitation</i>	1742	37.27	8.73	6.45	55.85
<i>Snowfall</i>	1742	33.44	25.08	0	118.1
<i>Number of Freezing Months</i> per Year	1742	3.26	1.58	0	6.27
<i>Distance</i> (in hundreds of miles)	1742	5.57	3.63	0.20	14.99
<i>Slot</i>	1742	0.20	0.40	0	1
<i>Population at Larger Endpoint</i> (in thousands)	1742	6744.24	5811.34	89.34	20196.65
<i>Population at Smaller Endpoint</i> (in thousands)	1742	1639.23	1592.54	63.16	8885.92

How To Model Choice Problem

- Are the unobservable determinants of “utility” correlated across options?
 - If yes, IIA property (“proportional substitution”) violated.
- Nested logit: IIA holds within nests but need not hold across nests
 - Good summary: Train’s book chapter 4
<http://elsa.berkeley.edu/books/choice2.html>



Bottom Nest: Own or Independent?

- Question: Why not use actual delay data?
- *Hub*
 - 60% of passengers are “connecting”
- *Number of Flights at Larger Endpoint*
- *Number of Flights at Smaller Endpoint*
- Max across 2 cities of:
 - *Precipitation*
 - *Snowfall*
 - *Number of Freezing Months per Year*
- *Slot constrained airport (ORD, JFK, LGA, DCA)*
- *Distance*
 - Longer flights have larger labor cost savings of indep

Top Nest: Use Regional?

- Expected Demand:
 - *Population at Larger Endpoint*
 - *Population at Smaller Endpoint*
 - *Hub* (higher demand?)
- *Distance*
 - Long = less pleasant & less fuel efficient
- *Slot*
 - Small planes less profitable if fewer landing slots

Table 4
Nested Logit

	(4-1)	(4-2)	(4-3)	(4-4)
Panel A: Bottom Nest	<i>Dependent Variable = 1 if Owned Regional is Used</i>			
<i>Hub</i>	1.429 (0.205)**	1.012 (0.280)**	3.527 (0.381)**	3.571 (0.440)**
<i>Precipitation</i>	0.022 (0.005)**	0.004 (0.005)	0.016 (0.006)*	0.002 (0.006)
<i>Snowfall</i>	0.020 (0.005)**	0.021 (0.005)**	0.024 (0.006)**	0.021 (0.006)**
<i>Number of Freezing Months per Year</i>	-0.711 (0.087)**	-0.737 (0.091)**	-0.756 (0.123)**	-0.735 (0.124)**
<i>Distance</i>	0.030 (0.040)	0.035 (0.040)	-0.122 (0.063)+	-0.090 (0.062)
<i>Slot</i>	0.640 (0.222)**	0.600 (0.225)*	1.398 (0.370)**	1.206 (0.384)**
<i>Number of Flights at Larger Endpoint</i>		0.230 (0.063)**		0.025 (0.072)
<i>Number of Flights at Smaller Endpoint</i>		2.229 (0.457)**		2.387 (0.548)**
Panel B: Top Nest	<i>Dependent Variable = 1 if Any Regional is Used</i>			
<i>Hub</i>	0.063 (0.162)	0.279 (0.169)	1.055 (0.385)**	1.526 (0.395)**
<i>ln(Population at Larger Endpoint)</i>	0.869 (0.064)**	0.850 (0.065)**	0.810 (0.081)**	0.747 (0.082)**
<i>ln(Population at Smaller Endpoint)</i>	-0.724 (0.071)**	-0.672 (0.073)**	-0.615 (0.089)**	-0.513 (0.092)**
<i>Distance</i>	-0.434 (0.025)**	-0.441 (0.028)**	-0.513 (0.042)**	-0.529 (0.045)**
<i>Slot</i>	-0.860 (0.174)**	-0.770 (0.191)**	-0.395 (0.300)	-0.082 (0.326)
Inclusive Values				
<i>Regional</i>	-0.250 (0.130)+	-0.508 (0.111)**	-0.603 (0.163)**	-0.738 (0.164)**
Carrier dummies	No	No	Yes	Yes
Observations	1742	1742	1159	1159

Standard errors in parentheses. + significant at 10 percent; * significant at 5 percent; ** significant at 1 percent. (1) and (2) include city pairs between top 300 U.S. airports operated by major network carriers (or their regional partners) in the 2nd quarter of 2000. (3) and (4) include only Continental, Delta, Northwest or US Airways (or their regional partners). *Precipitation* and *Snowfall* are measured in inches. *Distance* is measured in hundreds of miles. *Number of Flights at Larger (Smaller) Endpoint* is measured in hundreds.

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Hub ↑ 57.4 percentage points

0.1 more inches ↑ 0.11 percentage points

1.25 more inches (0.1 inch of water equivalent)
↑ 1.27 percentage points

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?!?! Cold weather reduces pr(owned)??

They investigate further in appendix. Cold is associated with *fewer* flight delays (while precip and snowfall have expected sign). Guess = cold, clear days

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Hub still significant (even after controlling for # flights at endpoint)

Top Nest

Panel B: Top Nest		<i>Dependent Variable = 1 if Any Regional is Used</i>			
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➔ Regionals used for small cities (at one endpoint), shorter routes, where the (large) airport is not slot-constrained

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Subsample
of majors
that use both
own &
independent

Control for
carrier
effects

Table 5
Nested Logit (All Carriers)
Hub Routes vs. Non-Hub Routes

	(5-1) Hub Routes	(5-2) Hub Routes	(5-3) Non-Hub Routes	(5-4) Non-Hub Routes
Panel A: Bottom Nest <i>Dependent Variable=1 if Owned Regional is Used</i>				
<i>Precipitation</i>	0.087 (0.009)**	0.058 (0.010)**	-0.018 (0.003)**	-0.010 (0.008)
<i>Snowfall</i>	0.021 (0.007)**	0.027 (0.007)**	0.001 (0.001)	0.022 (0.008)**
<i>Number of Freezing Months per Year</i>	-0.891 (0.120)**	-0.980 (0.124)**	-0.026 (0.024)	-0.488 (0.159)**
<i>Distance</i>	-0.011 (0.050)	-0.004 (0.049)	0.181 (0.039)**	-0.117 (0.084)
<i>Slot</i>	0.363 (0.318)	0.314 (0.327)	0.150 (0.231)	0.254 (0.358)
<i>Number of Flights at Larger Endpoint</i>		0.186 (0.072)*		1.371 (0.315)**
<i>Number of Flights at Smaller Endpoint</i>		5.472 (1.202)**		-1.076 (0.677)
Panel B: Top Nest <i>Dependent Variable=1 if Any Regional is Used</i>				
<i>ln(Population at Larger Endpoint)</i>	0.828 (0.070)**	0.817 (0.071)**	1.337 (0.206)**	0.780 (0.172)**
<i>ln(Population at Smaller Endpoint)</i>	-0.699 (0.081)**	-0.619 (0.084)**	-0.688 (0.173)**	-0.722 (0.167)**
<i>Distance</i>	-0.422 (0.028)**	-0.435 (0.030)**	-0.526 (0.348)	-0.478 (0.066)**
<i>Slot</i>	-1.447 (0.222)**	-1.514 (0.232)**	0.042 (1.447)	0.013 (0.344)
Inclusive Values				
<i>Regional</i>	-0.064 (0.073)	-0.264 (0.060)**	-11.836 (2.506)**	0.955 (0.670)
Observations	1317	1317	425	425

Separately for
hubs (75%)
vs. non-hubs (25%)

Even for hubs,
network
integration
matters

Other Interpretations

- Is *Snowfall* picking up unionization?
 - Robust to including dummies for 22 “right to work” states
- Is the “unit of choice” really an airport rather than a city-pair
 - E.g. For Atlanta, if Delta uses ASA (fully owned), it will use fully owned for all regional flights into Atlanta

Table 6B
Patterns of Regional Usage, Airport Analysis
Majors with Both Types of Regionals

	All Airports	Hubs	“Large” Airports	“Small” Airports
Number of observations	276	13	26	250
Number at which major uses both types of regional	73 (26.45 percent)	4 (30.77 percent)	13 (50 percent)	60 (24 percent)

Level of observation is the airline-airport. Sample includes airports at which Continental, Delta, Northwest or US Airways serves more than one regional route.

Additional questions about the data:

- What are the key variables in the empirical model?
- What is assumed to be exogenous and endogenous? Are there variables that are treated as exogenous that can conceivably be endogenous?
- What is the key variation in the data that is identifying the parameters of interest?
- Are there distributional assumptions that are driving identification?